



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

21-08-2021

JEE(Main + Advanced) : Nurture Course (PHASE : TEPS, TOPS, I, I(A), I(B), II(A) & NURTURE (LIVE) I, I(B), II(A))

ANSWER KEY**PAPER-1****PART-1 : PHYSICS**

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	C	B	B	A	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,B,C	A,B,C,D	A,D	A,C,D	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	6.00	2.00	4.00	2.00	6.00	0.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	C	C	C	D	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C,D	A,C,D	B,C	A,B,C,D	B,D	A,B
SECTION-II	Q.	1	2	3	4	5	6
	A.	15.00	3.00	11.00	24.00	0.25	25.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	C	A	C	A	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,B,C,D	A,B,D	A,D	A,C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	18.00	11.20	2.50	2.00	4.00	8.00

HINT – SHEET**PART-1 : PHYSICS****SECTION-I (i)**1. **Ans (B)**

By work energy theorem

$$\frac{1}{2}kx^2 - mg(h+x) = 0 - 0$$

2. **Ans (C)**

$$u = -kxy$$

$$\frac{\partial u}{\partial x} = -ky$$

$$\frac{\partial u}{\partial y} = -kx$$

$$F = - \left(\frac{\partial u}{\partial x} \hat{i} + \frac{\partial u}{\partial y} \hat{j} \right)$$

$$F = ky\hat{i} + kx\hat{j}$$

3. **Ans (B)**

$$v = t^2$$

$$\Rightarrow v_i = 0 ; v_f = 4$$

by work energy theorem

$$W_F = \Delta k$$

$$= \frac{1}{2} \cdot 2.16 = 16J$$

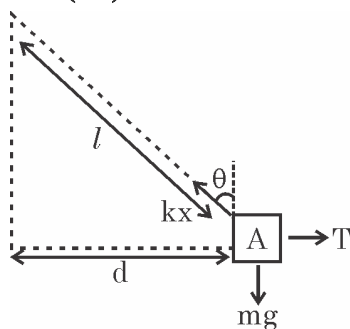
4. **Ans (B)**for eq. distance $\Rightarrow F_{net} = 0$

$$F = -\frac{du}{dx} = 0$$

$$-\frac{12a}{r^{13}} + \frac{6b}{r^7} = 0$$

$$r = \left(\frac{2a}{b} \right)^{1/6}$$

5. Ans (A)



$$kx \cos \theta = mg$$

$$x = \ell - 0.4 = \frac{0.4}{\cos \theta} - 0.4$$

$$\therefore k \left(\frac{0.4}{\cos \theta} - 0.4 \right) \cos \theta = mg$$

$$\cos \theta = \frac{1}{5}$$

$$\Rightarrow x = \frac{0.4 \times 5}{4} - 0.4 = 0.1 \text{ m}$$

$$\text{so } d = 0.4 \tan \theta = 0.3 \text{ m}$$

$$mgd - \frac{1}{2} kx^2 = \left(\frac{1}{2} mv^2 \right) \times 2$$

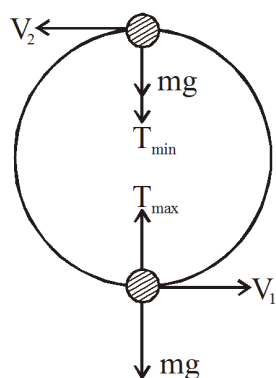
6. Ans (C)

$$\frac{1}{2} mv_1^2 = mg2\ell + \frac{1}{2} mv_2^2$$

$$\Rightarrow V_2^2 = V_1^2 - 4g\ell$$

$$T_{\max} = mg + \frac{mV_1^2}{\ell} \quad \dots(i)$$

$$T_{\min} + mg = \frac{mV_2^2}{\ell} \Rightarrow T_{\min} = \frac{mV_2^2}{\ell} - mg \quad \dots(ii)$$



Dividing eqⁿ (i) and eqⁿ (ii)

$$2 = \frac{V_1^2 + g\ell}{V_2^2 - g\ell} \Rightarrow 2V_2^2 - 2g\ell = V_1^2 + g\ell$$

Putting value of V_2^2

$$2(V_1^2 - 4g\ell) - 2g\ell = V_1^2 + g\ell$$

$$2V_1^2 - 8g\ell - 2g\ell = V_1^2 + g\ell \Rightarrow V_1 = \sqrt{11g\ell}$$

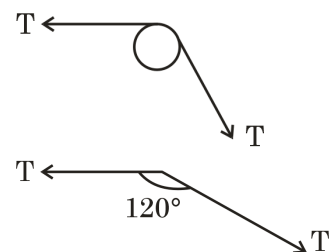
SECTION-I (ii)

7. Ans (A,B,C)

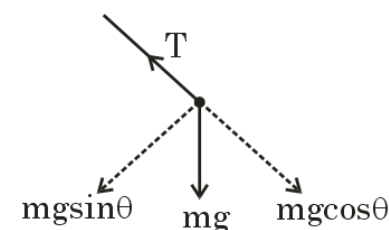
This system will act as rod rotating in circle.

$$\text{So } v_{\min} \text{ for complete circle} = \sqrt{4gR}$$

8. Ans (A,B,C)



$$F_{\text{net}} = T$$



$$T = mg \cos \theta = 100 \times \frac{\sqrt{3}}{2} = 50\sqrt{3}$$

$$mg \sin \theta = ma_t$$

$$\Rightarrow a_t = g \sin \theta = 5$$

9. Ans (A,B,C,D)

$$\text{for equation } F = -\frac{du}{dx} = 0$$

we get $x = 2$ & $x = 3$

$$\frac{d^2u}{dx^2} = 2x - 5$$

$$\text{at } x = 2, \frac{d^2u}{dx^2} < 0 \rightarrow \text{unstable}$$

$$\text{at } x = 3, \frac{d^2u}{dx^2} > 0 \rightarrow \text{stable}$$

No point of neutral equation

11. Ans (A,C,D)

Ans. (A,C,D)

$$W_{\text{mg}} = mgh = 5 \times 10 \times 6 = 300 \text{ J}$$

$$\Delta U = -300 \text{ J}$$

$W_N = 0$ (displacement is perpendicular to normal)

$$W_f = -\mu mg \cos \theta x = -0.5 \times 5 \times 10 \times \frac{4}{5} \times 10$$

$$= -200 \text{ J}$$

$$W_{\text{all}} = \Delta KE$$

12. Ans (A,C)

$$F = -\frac{\partial u}{\partial x}$$

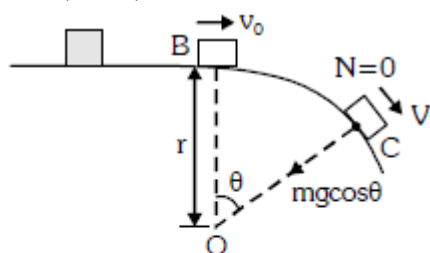
$$\text{At point B} \Rightarrow \frac{\partial u}{\partial x} < 0 \Rightarrow F > 0$$

$$\text{At point C} \frac{\partial u}{\partial x} = 0 \Rightarrow F = 0$$

$$\text{At equilibrium} \Rightarrow \frac{\partial u}{\partial x} = 0$$

SECTION-II

1. Ans (6.00)



$$-\Delta U = \Delta K$$

$$mg r (1 - \cos \theta) = \frac{1}{2} m(v^2 - v_0^2)$$

....(1)

$$mg \cos \theta = \frac{mv^2}{r}$$

2. Ans (2.00)

$$t = \sqrt{\frac{2H}{g}} = 2 \text{ sec}$$

3. Ans (4.00)

$$V = \sqrt{2g(0.8)}$$

$$V = 4 \text{ m/s}$$

4. Ans (2.00)

$$mgh = \frac{1}{2} kx^2$$

5. Ans (6.00)

By work energy theorem

$$-mg \times 2\ell = \frac{1}{2} mv_f^2 - \frac{1}{2} mv_i^2$$

$$v_f^2 - 10g\ell = -4g\ell$$

$$v_f = \sqrt{6g\ell}$$

$$x = 6$$

6. Ans (0.00)

$$\frac{du}{dx} = 0 = -16x + 4x^3$$

$$x(-4^2 + x^2) = 0$$

$$x = -2, 0, 2$$

$$\frac{d^2u}{dx^2} = -16 + 12x$$

$$\text{for } x = 0$$

$$\frac{d^2u}{dx^2} < 0 \text{ hence } x = 0 \text{ is Ans.}$$

PART-2 : CHEMISTRY

SECTION-I (i)

4. Ans (C)

$$\text{Molar mass of BeCl}_2 = 9 + 71 = 80 \text{ g/mol}$$

$$\Rightarrow \text{Moles of BeCl}_2 = \frac{0.8}{80} = 0.01$$

$$\therefore \text{Molarity} =$$

$$\frac{\text{Moles of BeCl}_2}{\text{Volume of solution (mL)}} \times 1000$$

$$= \frac{0.01}{500} \times 1000 = 0.02 \text{ M}$$

Now, take 20 mL of 0.02 M and dilute it 10 times

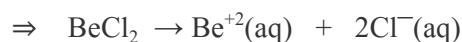
$$\text{So final volume} = 20 \times 10 = 200 \text{ mL}$$

Hence final molarity of this solution can be calculated as

$$\therefore M_1 V_1 = M_2 V_2$$

$$\Rightarrow 0.02 \times 20 = 200 \times M_2 \text{ (final molarity)}$$

$$\Rightarrow \boxed{M_2 = 2 \times 10^{-3} \text{ M}}$$



$$2 \times 10^{-3} \text{ M} \quad - \quad -$$

$$- \quad 2 \times 10^{-3} \text{ M} \quad (2 \times 2 \times 10^{-3}) \text{ M}$$

$$[\text{Cl}^{-}] = 4 \times 10^{-3} \text{ M}$$

$$\text{In 2 mL solution} \left(\frac{2}{1000} \text{ L solution} \right)$$

$$\therefore \text{Moles of Cl}^{-} = \text{Molarity} \times \text{volume in litre}$$

$$= 4 \times 10^{-3} \times \frac{2}{1000}$$

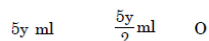
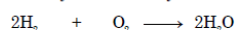
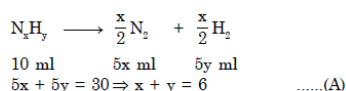
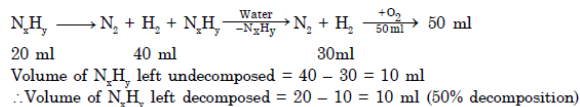
$$= 8 \times 10^{-6} \text{ moles}$$

6. Ans (B)

Class notes.

SECTION-I (ii)

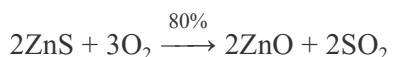
9. Ans (B,C)



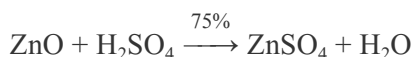
$$\text{Contraction in volume} : \frac{15y}{2} = 30 \Rightarrow y = 4, x = 2$$

$$\therefore \text{Compound} = \text{N}_2\text{H}_4$$

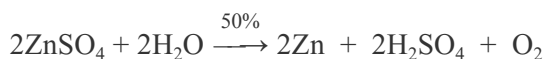
11. Ans (B,D)



$$15 \quad 15 \times 0.8 = 12$$



$$12 \quad 12 \times 0.75 = 9$$

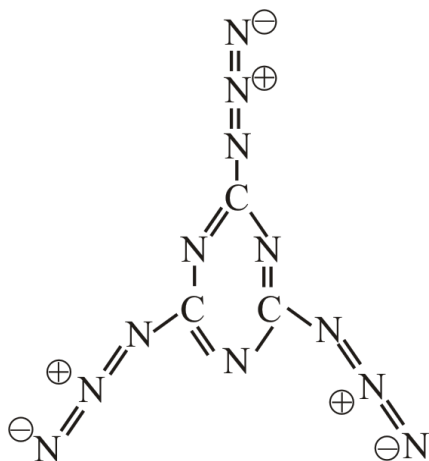


$$9 \quad 9 \times 0.5 \quad 9 \times 0.5 \quad 4.5 \times 0.5$$

$$= 4.5 \quad = 4.5 \quad = 2.25$$

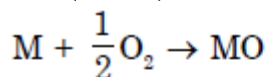
SECTION-II

1. Ans (15.00)



15 sigma bonds

4. Ans (24.00)

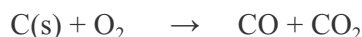


$$\text{Mole of O}_2 \text{ used} = \frac{8}{32} = 0.25 \text{ mole}$$

$$\text{Moles of metal reacted} = 2 \times n_{\text{O}_2} = 2 \times 0.25 = 0.5 \text{ mole}$$

$$\therefore \text{At. weight of metal} = \frac{12}{0.5} = 24$$

5. Ans (0.25)



$$2\text{mol} \quad 1.2\text{mol} \quad a \quad b$$

$$\text{POAC on C} \quad a + b = 2$$

$$\text{POAC on O} \quad a + 2b = 2 \times 1.2$$

$$b = 0.4 \quad a = 1.6$$

$$\frac{\text{Mole of CO}_2}{\text{Mole of CO}} = \frac{0.4}{1.6} = \frac{1}{4} = 0.25$$

6. Ans (25.00)

4th excited state i.e. in 5th shell

$$5s \quad 5p \quad 5d \quad 5f$$

$$1 + 9 + 7 + 5 + 3 = 25$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)

$$16\Delta^2 = a^2b^2\lambda$$

$$\Rightarrow 16 \left[\frac{1}{2} ab \sin c \right]^2 = a^2b^2\lambda$$

$$\Rightarrow 4\sin^2 c = \lambda$$

$$\therefore 0 < \lambda < 4$$

2. Ans (C)

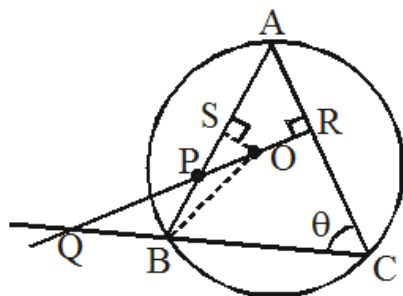
$$6, 7, c$$

$$c^2 + 6^2 > 7^2 \quad 6^2 + 7^2 > c^2$$

$$c^2 > 13 \quad c^2 < 85$$

$$\text{so } c = 4, 5, 6, 7, 8, 9$$

3. Ans (A)



Let $\angle \text{SOB} = \theta$

$$\Rightarrow \angle SBO = \angle PBO$$

$$= 90 - \theta$$

also $\angle RQC = 90 - \theta$

so $1 : 1$

4. Ans (C)

$$\frac{1}{2}\alpha a = \Delta \Rightarrow \alpha = \frac{2\Delta}{a}, \beta = \frac{2\Delta}{b}, \gamma = \frac{2\Delta}{c}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{a + b + c}{2\Delta} = \frac{1}{r}$$

$$r = \frac{\alpha\beta\gamma}{\alpha\beta + \beta\gamma + \gamma\alpha} = \frac{20}{44} = \frac{5}{11}$$

5. Ans (A)

$$D_r = r^2 \begin{vmatrix} 1 & 2021 & 2022 \\ 11 & 3 & 1 \\ 1 & \frac{1}{r} & \frac{1}{r^2} \end{vmatrix}$$

$$D_r = \begin{vmatrix} 1 & 2021 & 2022 \\ 11 & 3 & 1 \\ r^2 & r & 1 \end{vmatrix}$$

$\sum_{r=1}^5 D_r =$	1	2021	2022
	11	3	1
	55	15	5

$$= 0 \text{ (R}_2 \text{ \& R}_3 \text{ are proportional)}$$

6. Ans (D)

Given equation is $\tan(3x - 2x) = 1$

$$\Rightarrow \tan x = 1 \Rightarrow x = n\pi + \frac{\pi}{4}$$

but at this value $\tan 2x$ is undefined, hence there is no solution.

SECTION-I (ii)

7. Ans (A,B,C)

Let sides of triangle be $a-1, a, a+1$

$$S = \frac{3a}{2}$$

$$\Delta = \sqrt{\frac{3a}{2} \left(\frac{3a}{2} - a + 1 \right) \left(\frac{3a}{2} - a \right) \left(\frac{3a}{2} - a - 1 \right)}$$

$$= \frac{a}{4} \sqrt{3(a^2 - 4)}$$

$$R = \frac{abc}{4\Delta} = \frac{8}{\sqrt{15}} = \frac{a(a^2 - 1)}{4\frac{a}{4}\sqrt{3(a^2 - 4)}}$$

$$\Rightarrow \frac{8}{\sqrt{15}} = \frac{a^2 - 1}{\sqrt{3(a^2 - 4)}}$$

on solving we get $a = 3$

8. Ans (A,B,C,D)

$$(A) \sin(180^\circ + 45^\circ) = -\frac{1}{\sqrt{2}}$$

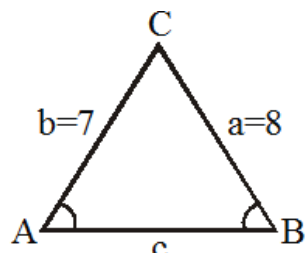
(B) $\sin 20^\circ \sin(60^\circ - 20^\circ) \sin(60^\circ + 20^\circ)$

$$= \frac{\sin 60^\circ}{4} = \frac{\sqrt{3}}{8}$$

(C) $(1 + \tan 3^\circ)(1 + \tan(45^\circ - 3^\circ)) = 2$

$$(D) \frac{2 \sin 3\theta \cos 2\theta}{2 \cos 3\theta \cos 2\theta} = \tan 3\theta$$

9. Ans (A,B,D)



$$\tan\left(\frac{A-B}{2}\right) = \left(\frac{a-b}{a+b}\right) \cot\left(\frac{C}{2}\right)$$

$$\Rightarrow \frac{1}{\sqrt{23}} = \frac{1}{15} \cot\left(\frac{C}{2}\right)$$

$$\Rightarrow \tan\left(\frac{C}{2}\right) = \frac{\sqrt{23}}{15}$$

$$\Rightarrow \sin C = \frac{2 \tan \frac{C}{2}}{1 + \tan^2 \frac{C}{2}}$$

$$\Rightarrow \sin C = \frac{15\sqrt{23}}{124}$$

$$\Delta = \frac{105\sqrt{23}}{31}$$

10. Ans (A,D)

$$\text{Expression} = \log_{10}(\tan 6^\circ) + \log_{10}(\tan 12^\circ) + \dots + \log_{10}(\tan 84^\circ)$$

$$= \log_{10}((\tan 6^\circ \cdot \tan 84^\circ) \cdot (\tan 12^\circ \cdot \tan 78^\circ) \dots (\tan 42^\circ \cdot \tan 48^\circ))$$

$$= \log_{10}((\tan 6^\circ \cdot \cot 6^\circ) \cdot (\tan 12^\circ \cdot \cot 12^\circ) \dots (\tan 42^\circ \cdot \cot 42^\circ))$$

$$= \log_{10} 1 = 0$$

11. Ans (A,C,D)

$$x^3 + 2x^2 - 2x - 1 = (x-1)(x^2 + 3x + 1)$$

$$x^2 + 3x + 1 = 0$$

$$ax^2 + bx + a = 0$$

$$a = 1, b = 3.$$

12. Ans (A,B,C)

$$a^2 + 1 = \sin \theta \cos \theta + a \sin \theta + a \cos \theta$$

$$a^2 + \sin^2 \theta + \cos^2 \theta = \sin \theta \cos \theta + a \sin \theta + a \cos \theta$$

$$\Rightarrow \sin \theta = \cos \theta = a$$

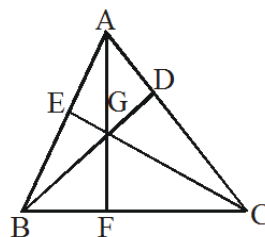
$$\Rightarrow \tan \theta = 1 \text{ \& } a = \frac{1}{\sqrt{2}} \text{ or } a = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow \text{sum of possible values of 'a' is zero.}$$

$$\tan \theta = 1 \text{ has 4 solution in } [0, 4\pi].$$

SECTION-II

1. Ans (18.00)



$$BD = \frac{9}{2} \Rightarrow BG = \frac{2}{3}BD = 3$$

$$CE = 6 \Rightarrow CG = \frac{2}{3}CE = 4$$

$$AF = \frac{15}{2} \Rightarrow GF = \frac{1}{3}AF = \frac{5}{2}$$

$$\text{Let } BC = a$$

$$GF = \frac{1}{2} \sqrt{2(BG)^2 + 2(CG)^2 - (BC)^2}$$

$$25 = 18 + 32 - a^2 \Rightarrow a = 5$$

$$\text{Now, } BG^2 + GC^2 = BC^2$$

$$\therefore \Delta BGC \text{ is right angled triangle}$$

$$\text{area of } \Delta ABC = 3\Delta BGC = 3 \times \frac{1}{2} \times 3 \times 4 = 18$$

2. Ans (11.20)

$$P = (\sqrt[5]{7})^{10} = (7)^{\frac{10}{5}} = 49$$

$$Q = 6 + \left(\frac{5}{2}\right)^{\log_9 8}$$

$$= 6 + \left(\frac{5}{2}\right)^2 = 6 + \frac{25}{4} = \frac{49}{4}$$

$$R = (8)^3 - (8)^2 = 448$$

$$\frac{QR}{10P} = \frac{49 \times 448}{10 \times 4 \times 49} = 11.20$$

3. Ans (2.50)

$$\frac{\cos A}{2R \sin A} = \frac{2 \cos B}{2R \sin B} = \frac{3 \cos C}{2R \sin C}$$

$$\Rightarrow \tan A = \frac{\tan B}{2} = \frac{\tan C}{3}$$

$$\because \Sigma \tan A = \Pi \tan A \Rightarrow \tan A=1, \tan B=2, \tan C=3$$

$$\begin{aligned} \Rightarrow \frac{b}{c} \sin 2A &= \frac{\sin B}{\sin C} \left(\frac{2 \tan A}{1 + \tan^2 A} \right) \\ &= \frac{2\sqrt{10}}{\sqrt{5} \cdot 3} \cdot \frac{2}{2} = \frac{2\sqrt{2}}{3} \end{aligned}$$

4. Ans (2.00)

$$\begin{vmatrix} 1 & \cos \alpha - \sin \alpha & \cos \alpha + \sin \alpha \\ 1 & \cos \beta - \sin \beta & \cos \beta + \sin \beta \\ 1 & \cos \gamma - \sin \gamma & \cos \gamma + \sin \gamma \end{vmatrix}$$

$$C_2 \rightarrow C_2 + C_3$$

$$= 2 \begin{vmatrix} 1 & \cos \alpha & \cos \alpha + \sin \alpha \\ 1 & \cos \beta & \cos \beta + \sin \beta \\ 1 & \cos \gamma & \cos \gamma + \sin \gamma \end{vmatrix}$$

$$C_3 \rightarrow C_3 - C_2$$

$$= 2 \begin{vmatrix} 1 & \cos \alpha & \sin \alpha \\ 1 & \cos \beta & \sin \beta \\ 1 & \cos \gamma & \sin \gamma \end{vmatrix} \Rightarrow k = 2$$

5. Ans (4.00)

$$xz = y^2 \text{ \& } y^2 = \frac{x+z}{2}$$

$$\Rightarrow \frac{x+z}{2} = xz$$

$$A.M. \geq G.M.$$

$$\Rightarrow xz \geq \sqrt{xz} \Rightarrow (xz)^2 \geq xz$$

$$\because xz \neq 0 \Rightarrow xz \geq 1$$

$$\Rightarrow y^2 \geq 1 \Rightarrow y \geq 1$$

$$\text{Now, } x + 2y + z = 2y^2 + 2y$$

$$\therefore (2y^2 + 2y)_{\min} = 2 + 2 = 4$$

6. Ans (8.00)

$$\begin{aligned} N &= \frac{(10^{16} - 1) - \frac{5}{9}(10^8 - 1)}{9(10)^8 + 4} \\ &= \frac{(10^8 - 1) \left\{ (10^8 + 1) - \frac{5}{9} \right\}}{9(10)^8 + 4} = \frac{(10^8 - 1) \left(10^8 + \frac{4}{9} \right)}{9(10)^8 + 4} \\ &= \frac{(10^8 - 1) (9(10)^8 + 4)}{9 (9(10)^8 + 4)} \\ &= \frac{10^8 - 1}{9} = \underbrace{111 \dots 1}_{8 \text{ digits}} \end{aligned}$$



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ANSWER KEY**PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,C	B,C,D	A,C	B,D	A,B,C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	24.00	22.00	140.00	30.00	30.00	12.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	9	4	3	0	8

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,D	B,D	A,B,C,D	A,B,D	A,C	A,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	10.00	780.00	4.00	9.00	3.20	4.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	3	6	6	3	4

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C,D	B,D	A,B,C,D	A,B,C	B,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	81.60	3.00	4.50	1.00	0.50	4.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	6	7	6	6	6	2

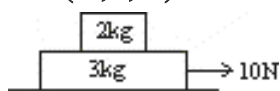
HINT – SHEET**PART-1 : PHYSICS****SECTION-I**

1. Ans (A,B,C)

$$\frac{1}{2}kx^2 = \frac{1}{2} \times 100 \times (0.5)^2 = 12.5J$$

$$W_{all} = \Delta KE$$

2. Ans (B,C,D)



$$a_{common} = \frac{10}{5} = 2m/sec^2$$

$$F_{req} \text{ to move} = 2 \times 2 = 4N$$

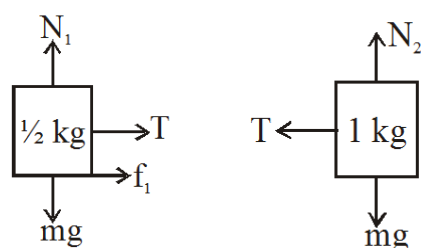
$$\mu N \geq 4$$

$$\mu \cdot 20 \geq 4$$

$$\mu \geq \frac{1}{5}$$

$$\mu \geq 0.2$$

3. Ans (A,C)



$$f_1 + T = m_1 \omega^2 r_1$$

$$f_1 + T = \frac{1}{2} \cdot 1.1$$

$$f_1 + T = \frac{1}{2}$$

Put value of T

$$f_1 = \frac{-3}{2}N$$

-ve shows opposite to what we have assume direction.

$$T = m^2 \omega^2 r_2$$

$$T = 1.1.2$$

$$T = 2N \quad \dots(i)$$

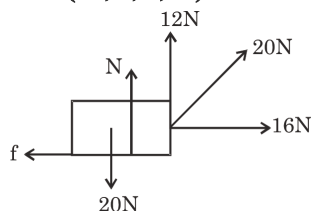
4. Ans (B,D)

$$\theta = t^2 - 6t + 8$$

$$\frac{d\theta}{dt} = \omega = 2t - 6 \Rightarrow \omega = 0 \text{ at } t = 3 \text{ sec}$$

$$\alpha = \frac{d\omega}{dt} = 2 = \text{constant.}$$

5. Ans (A,B,C,D)



$$N + 12 = 20 \Rightarrow N = 20 - 12 = 8 \text{ N}$$

$$f = \mu N = 0.5 \times (20 - 12) = 4 \text{ N}$$

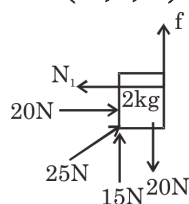
$$16 - 4 = 2a$$

$$a = 6 \text{ m/s}^2$$

$$W_F = 16 \times 2 = 32 \text{ J}$$

$$W_f = -4 \times 2 = -8 \text{ J}$$

6. Ans (A,B,D)



$$f + 15 = 20$$

$$f = 5 \text{ N}$$

$$N_1 - 20 = 0$$

$$N_1 = 20$$

SECTION-II

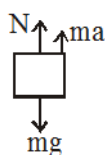
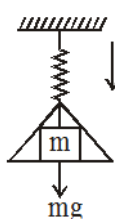
1. Ans (24.00)

Initial reading of spring balance = $mg = 49 \text{ N}$. when spring balance start moving downward with acceleration (5 m/s^2). So here normal reaction on spring balance plate will change due to pseudo force.

$$N = mg - ma$$

$$N = 49 - \left(\frac{49}{9.8} \times 5 \right)$$

$$N = 24 \text{ N}$$



2. Ans (22.00)

$$\text{Velocity at C} = \sqrt{2gr}$$

$$a_{\text{net}} = \sqrt{g^2 + (2g)^2} = \sqrt{5}g$$

3. Ans (140.00)

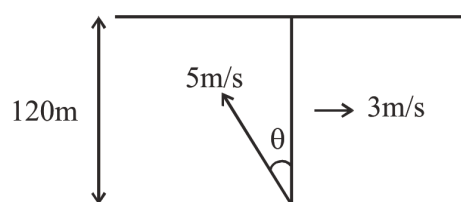
$$P = \vec{F} \cdot \vec{V}$$

$$P = 50 - 30 + 120$$

$$P = 140 \text{ Watt}$$

4. Ans (30.00)

For shortest path



$$120 = 5 \cos \theta \cdot t$$

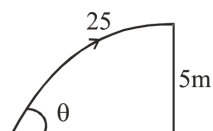
$$5 \sin \theta = 3$$

$$\sin \theta = \frac{3}{5} \Rightarrow \theta = 37^\circ$$

$$120 = 5 \times \frac{4}{5} \times t$$

$$t = 30 \text{ s}$$

5. Ans (30.00)



$$\text{in } t = 2 \text{ sec}$$

$$S_y = 5 \text{ m}$$

$$v_y = 25 \sin \theta$$

$$a_y = -g$$

$$5 = 25 \sin \theta \times 2 - \frac{1}{2} \times 10 \times (2)^2$$

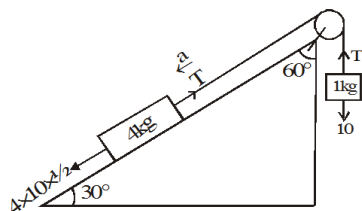
$$5 = 50 \sin \theta - 5 \times 4$$

$$5 = 50 \sin \theta - 25$$

$$25 = 50 \sin \theta$$

$$\sin \theta = \frac{1}{2} \quad \theta = 30^\circ$$

6. Ans (12.00)



$$20 - T = 4a \dots (1)$$

$$T - 10 = a \dots (2)$$

$$5a = 10$$

$$a = 2 \text{ m/sec}^2$$

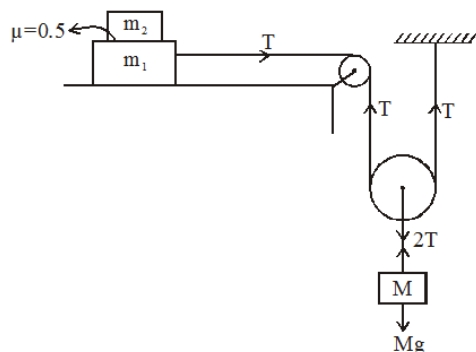
put in equation (i)

$$20 - T = 8$$

$$T = 12 \text{ N}$$

SECTION-III

1. Ans (3)



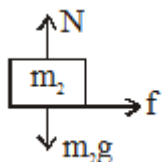
$$m_1 = 2 \text{ kg}$$

$$m_2 = 1 \text{ kg}$$

maximum friction force between block m_1 and m_2 is

$$f_{\max} = \mu N$$

$$= 0.5 \times 1 \times 10 = 5 \text{ N}$$



$$ma_{\max} = f_{\max}$$

$$a_{\max} = \frac{5}{1} = 5 \text{ m/sec}^2$$

$$\text{Now } T = ma$$

$$= (2 + 1) \times 5$$

$$= 15 \text{ N}$$

$$\text{Now } 2T = Mg$$

$$30 = M \times 10$$

$$M = 3 \text{ kg}$$

2. Ans (9)

Mechanical energy $E = K + U = \text{constant}$

$$\text{At } x = 5 \text{ m, } E = K + U = 20 + (5-2)^2 - 20 = 9 \text{ J}$$

$$\Rightarrow \text{At } x = 2 \text{ m, } E = 9 \text{ J}$$

3. Ans (4)

$$\langle F \rangle = m \langle a \rangle$$

$$= \frac{m [v_0 - (-v_0)]}{\pi R / v_0} = \frac{2mv_0^2}{\pi R}$$

4. Ans (3)

$$\mathbf{V}_{AB} = 24\hat{i} + 24\hat{j}$$

$$T = \frac{d}{24}$$

$$30 = 24T$$

5. Ans (0)

$$\frac{du}{dx} = 0 = -16x + 4x^3$$

$$x(-4^2 + x^2) = 0$$

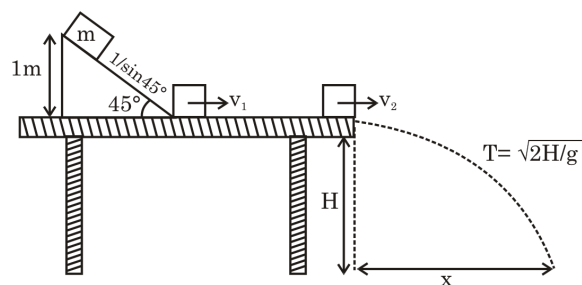
$$x = -4, 0, 4$$

$$\frac{d^2u}{dx^2} = -16 + 12x$$

$$\text{for } x = 0$$

$$\frac{d^2u}{dx^2} < 0 \text{ hence } x = 0 \text{ is Ans.}$$

6. Ans (8)



$$(v_1)^2 = 0^2 + 2g \sin 45^\circ \times \frac{1}{\sin 45}$$

$$(v_2)^2 = (v_1)^2 + 2(-\mu g) \times 1$$

$$v_2^2 = 2g - 2g \times 0.2 = 20 - 4 = 16$$

$$v_2^2 = 16 \Rightarrow v_2 = 4 \text{ m/s}$$

$$x = \left[\sqrt{\frac{2 \times 20}{g}} \right] \times 4 = 8 \text{ m}$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (B,D)**

J : Number of valence e^{-} 's = 5

K : Number of valence e^{-} 's = 4

L : Number of valence e^{-} 's > 7 (no sharp pump in the given data)

M : Number of valence e^{-} 's = 5

N : Number of valence e^{-} 's = 4

5. **Ans (A,C)**

Molar mass of H_2O_2 = 34 g/mol

($n_{H_2O_2}$ = 1 mole)

$$\therefore \text{Molarity} = \frac{\text{moles}}{\text{volume (L)}}$$

$$\therefore \text{Molarity} = \frac{1}{2} = 0.5$$

$\therefore$ At 1 atm, 273 K

$$\boxed{\text{Vol. strength} = \text{Molarity} \times 11.2}$$

$$= 0.5 \times 11.2$$

$$= 5.6 \text{ V } H_2O_2 \quad (A) \rightarrow \text{Correct}$$

(B) Density of solution is not given :

(Generally assume 1g/mol)

$$\therefore \text{mass of solution} = 2000 \text{ gm}$$

$$\therefore \text{mass of solvent} = (2000 - 34) = 1966 \text{ gm}$$

$$\therefore m = \frac{\text{moles of } H_2O_2}{\text{mass of solvent (gm)}} \times 1000$$

$$= \frac{1}{1966} \times 1000 = 0.5086 \text{ m (B)} \rightarrow \text{Incorrect}$$

$$(C) \% \frac{w}{v} = \frac{\text{wt. of } H_2O_2 \text{ (solute)}}{\text{volume of solution (mL)}} \times 100$$

$$= \frac{34}{2000} \times 100 = 1.7 \% \frac{w}{v}$$

(C) $\rightarrow$ correct

(D) Incorrect

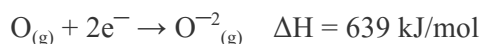
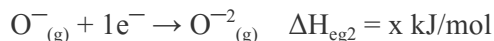
Because (volume of aqueous solution $\neq$ volume of gas)

volume of gas depends on P & T.

SECTION-II

2. **Ans (780.00)**

ΔH_{eg} (Electron gain enthalpy)



$$\Delta H = \Delta H_{eg1} + \Delta H_{eg2}$$

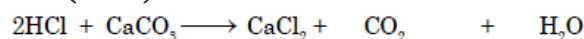
$$639 = -141 + x$$

$$\Rightarrow x = 780$$

$$\Rightarrow \Delta H_{eg2} = 780 \text{ kJ/mol}$$

Ans. (780)

3. **Ans (4.00)**



$$\frac{2.5 \text{ gm}}{62.5} = \frac{1.1}{44} = 0.025$$

$$\text{Mass} = \frac{2.5}{62.5} \times 100 = 4\%$$

4. **Ans (9.00)**

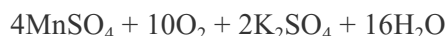
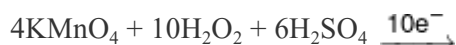
$$M = \frac{\%w/w \times d \times 10}{M_w} = \frac{49 \times 1.8 \times 10}{98} = 9M$$

SECTION-III

1. **Ans (3)**

($NF_3 < N(Me)_3$) ; ($SiCl_4, CCl_4$), (H_2, O_2)

3. **Ans (6)**



$$x = 4, z = 6, p = 4$$

4. **Ans (6)**



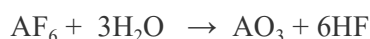
$$n\text{-factor} = 3 \times 2 \quad n\text{-factor} = 5$$

$$\text{Eq. of } Fe_2(C_2O_4)_3 = \text{Eq. of } KMnO_4$$

$$5 \times 6 = 5 \times n$$

$$n = 6$$

5. **Ans (3)**



6. Ans (4)

$O^{2-}, F^-, Na^+, Mg^{2+}, Al^{3+}$ (Isoelectronic)

—————→ Z ↑

e^- same, σ same

$Z_{\text{eff}} = (Z - \sigma) \therefore \text{size} \downarrow$

size $O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$

Down the group size increases

$\therefore Na^+ < K^+ < Rb^+$

$\therefore$ Ionic-radius greater than Na^+

$\Rightarrow (O^{2-}, F^-, K^+, Rb^+)$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (A,C,D)

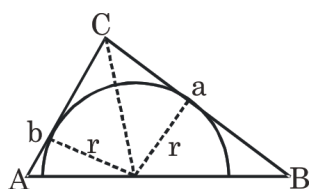
$$\sin(A + C) + \sin(A - C) = \sin B$$

$$\sin(\pi - B) + \sin(A - C) = \sin B$$

$$A = C$$

It is isosceles triangle

2. Ans (B,D)



$$\Delta = \frac{1}{2}a \cdot r + \frac{1}{2}b \cdot r \Rightarrow r = \frac{2\Delta}{a+b}$$

Also

$$\cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \sqrt{\frac{s(s-a)}{bc}} \cdot \sqrt{\frac{s(s-b)}{ac}} \cdot \sqrt{\frac{s(s-c)}{ab}}$$

$$= \frac{s\Delta}{abc}$$

$$\Rightarrow \frac{2abc}{s(a+b)} \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \frac{2abc}{s(a+b)} \times \frac{s\Delta}{abc}$$

$$= \frac{2\Delta}{a+b} = r$$

3. Ans (A,B,C,D)

$$C_1 \rightarrow C_1 - C_2, C_2 \rightarrow C_2 - C_3$$

$$\begin{vmatrix} 1 & 4 & r-6 \\ 2 & 5 & 2r-11 \\ 3 & 6 & 3r-18 \end{vmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 3R_1$$

$$\begin{vmatrix} 1 & 4 & r-6 \\ 0 & -3 & 1 \\ 0 & -6 & 0 \end{vmatrix} = 6$$

$$\Rightarrow T_r = 6$$

4. Ans (A,B,C)

$$\sin\left(\frac{\pi x^2}{3}\right) = 1$$

$$x^2 = \frac{3}{2}(4n+1); n \in \mathbb{I}$$

Only $n = 0$ and $n = 1$ are possible

$$\therefore x^2 = \frac{3}{2} \text{ or } x^2 = \frac{15}{2}$$

$$\text{Hence, } x = \pm\sqrt{\frac{3}{2}} \text{ or } \pm\sqrt{\frac{15}{2}}$$

For each value of x we will get 2 value of y ,

hence 8 ordered pairs.

5. Ans (B,D)

$$3\sin x + 4\cos x = (a+1)^2 + 5$$

$$a = -1 \text{ and } 3\sin x + 4\cos x = 5$$

$$\cos(x - \alpha) = 1 \quad \cos \alpha = \frac{4}{5}$$

$$x - \alpha = 0, 2\pi \quad x = \alpha, 2\pi + \alpha$$

6. Ans (A,B,D)

For A.P.

$$a = 4 ; b = 7$$

$$T_n = 11n - 8$$

$$T_1 = 3 ; T_2 = 14$$

$$d = 11$$

$$a^3 - b^2 = 64 - 49 = 15$$

$$S_8 = \frac{8}{2} [2 \times 3 + (8-1)11]$$

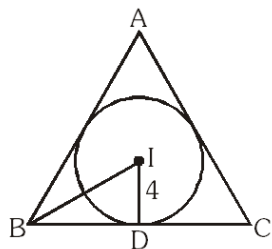
$$= 4[6 + 77] = 4 \times 83 = 332$$

SECTION-II

1. Ans (81.60)

Clearly, incentre is coincident with circumcentre.

Hence triangle is equilateral



Here $IB = R = 2r = 8$ (As $r = 4$)

$$\Rightarrow BD = \sqrt{8^2 - 4^2} = \sqrt{48} = 4\sqrt{3}$$

Hence $BC = 2BD = 8\sqrt{3}$

Hence Area of $\Delta ABC =$

$$\frac{\sqrt{3}}{4} a^2 = \frac{\sqrt{3}}{4} (8\sqrt{3})^2 = 48\sqrt{3}$$

2. Ans (3.00)

$$f(\theta) = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 + \sin \theta & 1 \\ 1 & 1 & 1 + \cos \theta \end{vmatrix}$$

Apply $C_1 \rightarrow C_1 - C_3 ; C_2 \rightarrow C_2 - C_3$

$$f(\theta) = \begin{vmatrix} 0 & 0 & 1 \\ 0 & \sin \theta & 1 \\ -\cos \theta & -\cos \theta & 1 + \cos \theta \end{vmatrix}$$

$$\Rightarrow f(\theta) = \sin \theta \cdot \cos \theta$$

$$\sin \theta \cdot \cos \theta = 0 \Rightarrow \theta = 0, \frac{\pi}{2}, \frac{3\pi}{2}$$

Three solutions.

3. Ans (4.50)

$$2 + 22 + 222 + 2222 + \dots 10 \text{ terms}$$

$$= 2[1 + 11 + 111 + 1111 + \dots 10 \text{ terms}]$$

$$= \frac{2}{9} [9 + 99 + 999 + 9999 + \dots 10 \text{ terms}]$$

$$= \frac{2}{9} [(10 - 1) + (10^2 - 1) + (10^3 - 1) + \dots]$$

...

$$= \frac{2}{9} [(10 + 10^2 + 10^3 + \dots 10 \text{ terms}) - (1 + 1 + \dots 10 \text{ terms})]$$

...

$$= \frac{2}{9} \left[10 \left(\frac{10^{10} - 1}{10 - 1} \right) - 10 \right] \quad \text{Compare}$$

$$= \frac{2}{9} \left[\frac{10^{11} - 10 - 90}{9} \right] \quad a = 2; c = 9$$

=

$$\frac{2(10^{11} - 10^2)}{(9)^2} \therefore \frac{c}{a} = \frac{9}{2} = 4.5$$

4. Ans (1.00)

$$x^2 - bx + c = 0 \begin{cases} \alpha \\ \beta \end{cases}$$

$$|\alpha - \beta| = 1 \Rightarrow \frac{\sqrt{b^2 - 4c}}{1} = 1$$

$$\Rightarrow b^2 - 4c = 1$$

5. Ans (0.50)

$$\frac{r}{R} = \frac{\Delta}{s \cdot \frac{abc}{4\Delta}} = \frac{4\Delta^2}{s \cdot abc}$$

$$= \frac{4(s-a)(s-b)(s-c)}{abc}$$

$$s = \frac{15\lambda}{2} \Rightarrow \frac{4 \left(\frac{15\lambda}{2} - 4\lambda \right) \left(\frac{15\lambda}{2} - 5\lambda \right) \left(\frac{15\lambda}{2} - 6\lambda \right)}{(4\lambda)(5\lambda)(6\lambda)}$$

$$= \frac{4 \cdot 7 \cdot 5 \cdot 3}{8 \cdot 4 \cdot 5 \cdot 6} = \frac{35}{80} = \frac{7}{16} \Rightarrow \frac{8r}{7R} = \frac{1}{2}$$

6. Ans (4.00)

$$(\sin^2 x - \cos^2 x) + \left(\frac{1}{\sin^2 x} - \frac{1}{\cos^2 x} \right) = 0$$

$$(\sin^2 x - \cos^2 x) - \frac{(\sin^2 x - \cos^2 x)}{\sin^2 x \cos^2 x} = 0$$

$$\sin^2 x = \cos^2 x \text{ or } \sin^2 x \cos^2 x = 1$$

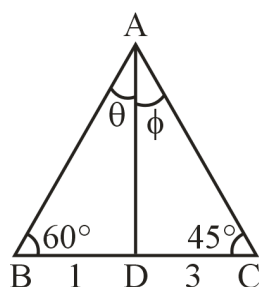
$$\sin^2 2x = 4 \text{ (Not Possible)}$$

$$\tan^2 x = 1$$

possible for four values in $[0, 2\pi]$

SECTION-III

1. Ans (6)



$$\angle A = 75^\circ, \angle B = 60^\circ, \angle C = 45^\circ$$

$$\text{Now } \frac{BD}{\sin(\angle BAD)} = \frac{AD}{\sin 60^\circ}$$

$$\Rightarrow \sin(\angle BAD) = \frac{BD}{AD} \times \frac{\sqrt{3}}{2} \dots(i)$$

$$\text{Also } \frac{CD}{\sin(\angle CAD)} = \frac{AD}{\sin 45^\circ}$$

$$\Rightarrow \sin(\angle CAD) = \frac{CD}{AD} \times \frac{1}{\sqrt{2}} \dots(ii)$$

Divide (i) by (ii)

$$\Rightarrow \frac{\sin(\angle BAD)}{\sin(\angle CAD)} = \frac{1}{\sqrt{6}}$$

2. Ans (7)

$$\text{Put } a = \log_3 \log_4 5 \text{ we get } f(x) = 16.5^{-2\cos x}$$

$$\text{given } 80 < f(x) < 400$$

$$\Rightarrow 5 < 5^{-2\cos x} < 25$$

$$\therefore x \in \left[\frac{2\pi}{3}, \frac{4\pi}{3} \right]$$

3. Ans (6)

Angles are A, 2A, 3A

$$\Rightarrow A = 30^\circ \quad \text{sides are}$$

$$B = 60^\circ \quad \sin A; \sin B; \sin C$$

$$C = 90^\circ \quad \frac{1}{2} : \frac{\sqrt{3}}{2} : 1$$

$$1 : \sqrt{3} : 2$$

4. Ans (6)

$$\frac{1 - \cos 2A}{2} + \frac{1 - \cos 2B}{2} + \frac{1 - \cos 2C}{2}$$

$$\Rightarrow \frac{3 - (-1 - 4\cos A \cos B \cos C)}{2}$$

$$\Rightarrow \frac{4}{2} = 2 \quad (\because \cos A \cos B \cos C = 0)$$

triangle is right angled

5. Ans (6)

$$|x - 3| = t$$

$$t^2 - t - 2 = 0 \Rightarrow |x - 3| = -1 \text{ not possible}$$

$$\text{or } |x - 3| = 2 \Rightarrow x - 3 = \pm 2$$

$$x = 5, 1$$

6. Ans (2)

$$\frac{1 + \log_{3b} 2b \times \log_{2b} b \times \log_{4b} 3b}{\log_{3b} 2b \times \log_{4b} 3b} = \frac{1 + \log_{4b} b}{\log_{4b} 2b} = \frac{\log_{4b} 4b^2}{\log_{4b} 2b} = 2$$